00110

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Cyber Security

Final Examination
Year 3, Semester 1&2 (2022)

IE3032 – Network Security

Duration: 2 Hours

June 2022

Instructions to Candidates:

- This paper has 4 Questions.
- Answer all questions in the booklet given.
- The total mark for the paper is 100.
- This paper contains 6 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Q1. IoT and Related Case Study [15 marks]

1. IoT networks comprises of following three layers: Perception Layer (Data gathering layer) Network Layer, and Analysis Layer.

   1.1. Discuss one important function by providing examples **(6 marks)**

2. "IoT cyberattacks more than doubled year-on-year during the first half of 2021, according to anti-virus and computer security service provider Kaspersky. From January to June this year, some 1.51 billion breaches of Internet of Things (IoT) devices took place, Kaspersky reported, an increase from 639 million in 2020. Most attackers brokered access to IoT networks via the telnet protocol, a command line interface that enables remote communication with a device or server. Kaspersky found that more than 872 million of IoT cyberattacks — or 58% – leveraged telnet, many with the intent of cryptocurrency mining, distributed denial-of-service (DDoS) shutdowns or pilfering confidential data. The study was conducted using software honeypots, which emulate IoT devices as a proxy for vulnerable hardware endpoints, according to PYMTS.com. The findings confirm that the pandemic has exacerbated IoT vulnerabilities by prolonging device usage in household settings. Many of these devices – whether intended for enterprise or personal use – lack adequate security protocols. The spike in IoT cyberattacks comes at a sensitive time for the industry. New technologies such as artificial intelligence and edge computing have presented opportunities, but also complicate the cyber and data security landscape. Dan Demeter, security expert at Kaspersky, said: "We see that once users' interest in smart devices rose, attacks also intensified."

   2.1. What are the common IoT attacks mentioned in the above report? **(3 marks)**

   2.2. Select one attack and explain the vulnerability leads to that attack. **(2 marks)**

   2.3. Create a solution to mitigate the attack and provide the technical measures that can be taken to avoid the vulnerability. **(4 marks)**

Q2. Defending the Network [35 marks]

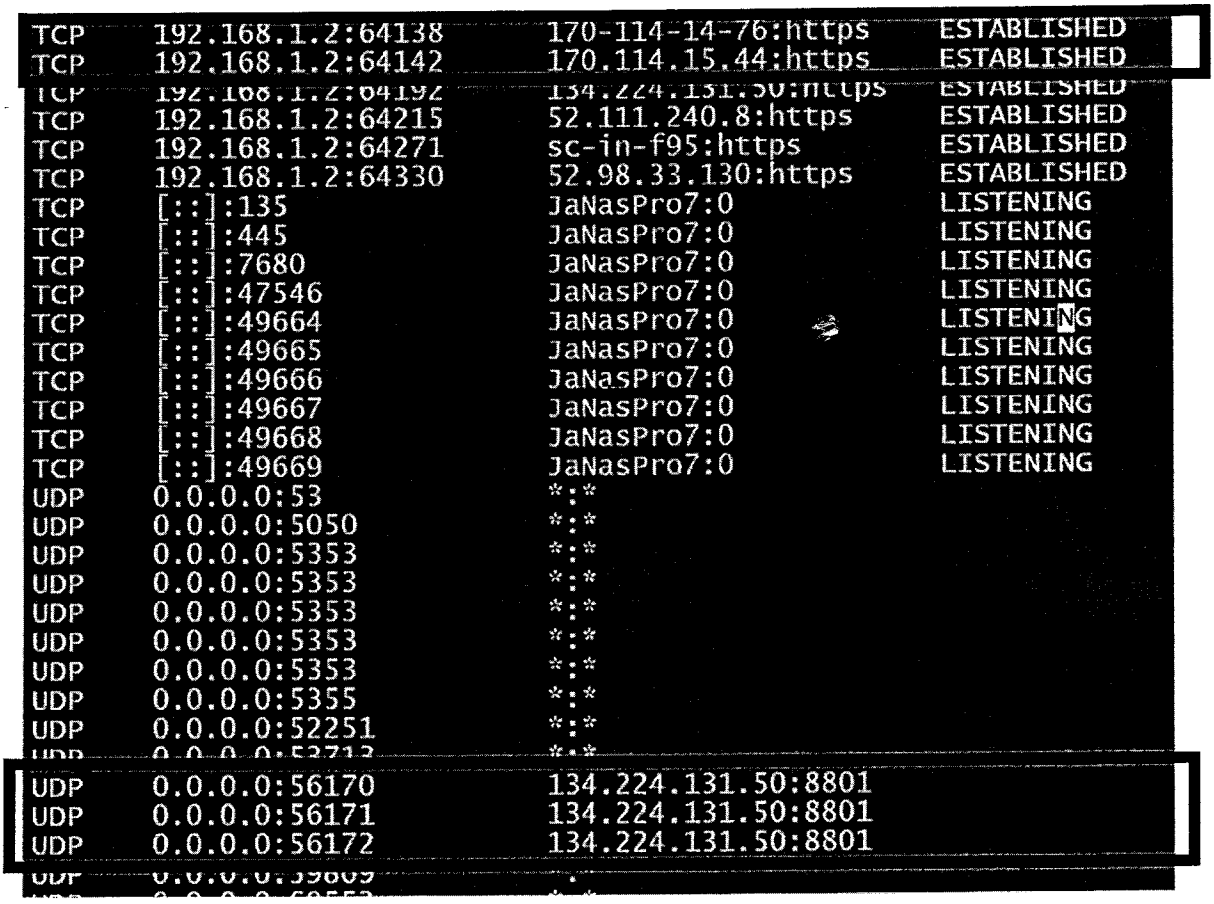
1. Explain what are the two categories of Endpoint security? Your answer should contain examples for each category **(4 marks)**
2. ISO-OSI reference model has 7 layers to ensure data delivery in computer networks. TCP/ IP stack has been derived using the ISO-OSI reference model for Internet communication.
   2.1. Explain about the DataLink Layer of ISO/OSI Ref. Model. Your answer should discuss about functionalities, addresses, security vulnerabilities, and attack **(4 marks)**
   2.2. **Illustrate** what is an Overlapping Fragmentation Attack? You have to clearly mention all the necessary steps in the figure. **(4 marks)**
   2.3. List 3 UDP attacks and explain one of the attacks. **(4 marks)**
3. A company is developing an online tutoring platform that allows registered students to seek help with their study and registered tutors to bid for a request. The platform is designed to work on both desktop computers and mobile devices. Students and tutors will be able to conduct one-on-one meeting and share their materials during a help session.
   3.1. Describe three (3) requirements of the online platform that address the security goal Integrity. Your answer must be specific to this scenario and cover all aspects of the security goal. **(3 Marks)**
   3.2. Due to the mal configuration of the password policy of the tutoring platform it was infected by "HermeticWizard" the data-wiper like a computer worm which is spreading across a local network.
      3.2.1. List five components that should be considered in password policy. **(5 Marks)**
      3.2.2. List down 4 phases that used when responding to a worm attack **(4 Marks)**
      3.2.3. Compare two network testing tools that could have been used to eliminate the worm attack to the tutoring platform. **(3 Marks)**
4. Use one example and explain what is Zero Trust **(4 Marks)**

Q3. Firewall and AAA [20 marks]

1. A consultant firm recommended the ACME company to implement an ACL to its network infrastructure to curb illegal activities. What is an ACL and name the 4 areas (i.e., ports or addresses) where ACLs can be applied to? **(4 Marks)**
2. "A firewall is a network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules."
   2.1. List 4 limitations of a Firewall and briefly explain them. **(3 Marks)**
   2.2. List 4 Firewall best practices that can be used to Secure Your Network **(3 Marks)**
3. Discuss why ACL and Stateful firewalls are not suitable to filter zoom traffic in a case of security threat. Hence, it is better we have a Next Generation firewall to filter the traffic. **(5 Marks)**
4. Compare Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access Control System (TACACS+). Your comparison should explain the functionalities, advantages and disadvantages. **(2 Marks)**
5. Single sign-on (SSO) is an authentication scheme that allows a user to log in with a single ID and password. Discuss the biggest disadvantage of using SSO for authentication and suggest a solution to overcome that disadvantage? **(3 Marks)**

Q4. TCP/IP Stack and the related attacks [30 marks]

1. Write 4 basic characteristics of reliable networks, and explain one of them using Zoom as an example **(5 marks)**
2. When using the Zoom communication to have online lectures, as indicated in figure below, we can observe that Zoom uses both TCP and UDP ports to do the communication. First, Zoom initiates the connection with the Zoom servers, and then send Audio, Video and Static content (screen share) during the zoom session. Below figure shows the connections made by Zoom when conducing a lecture.

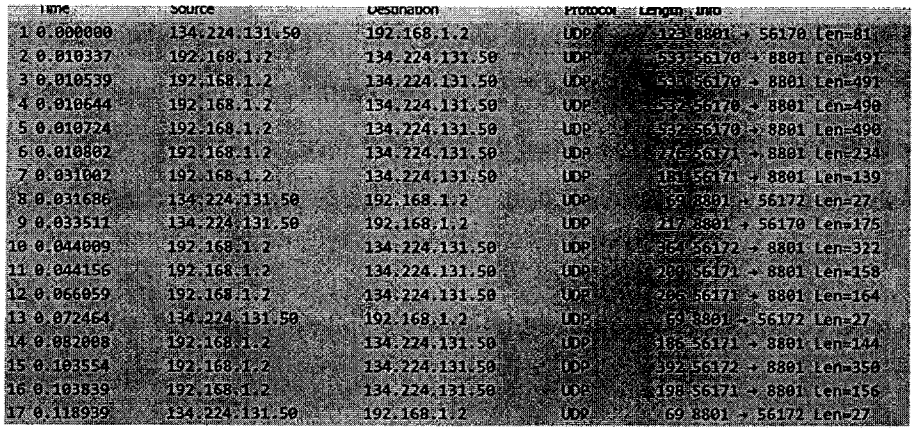


2.1. Analyze the above figure and explain why Zoom use both reliable and unreliable communication. Your explanation should have examples of services. **(6 marks)**

2.2. Above Figure displays three different UDP Source PORT numbers are assigned to the Zoom destination port.

2.2.1. What are those port numbers ? **(2 mark)**

2.2.2. Below Wireshark output shows the Zoom packet captured during the lecture. State which UDP port is used to send which data (Video, Audio and Static content)? You must justify your answer. **(5 marks)**



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	134.224.131.50	192.168.1.2	UDP	123	8801 → 56170 Len=81
2	0.010337	192.168.1.2	134.224.131.50	UDP	533	56170 → 8801 Len=491
3	0.010539	192.168.1.2	134.224.131.50	UDP	533	56170 → 8801 Len=491
4	0.010644	192.168.1.2	134.224.131.50	UDP	532	56170 → 8801 Len=490
5	0.010724	192.168.1.2	134.224.131.50	UDP	532	56170 → 8801 Len=490
6	0.010802	192.168.1.2	134.224.131.50	UDP	276	56171 → 8801 Len=234
7	0.031002	192.168.1.2	134.224.131.50	UDP	183	56171 → 8801 Len=139
8	0.031686	134.224.131.50	192.168.1.2	UDP	69	8801 → 56172 Len=27
9	0.033511	134.224.131.50	192.168.1.2	UDP	217	8801 → 56170 Len=175
10	0.044009	192.168.1.2	134.224.131.50	UDP	361	56172 → 8801 Len=322
11	0.044156	192.168.1.2	134.224.131.50	UDP	200	56171 → 8801 Len=158
12	0.066059	192.168.1.2	134.224.131.50	UDP	206	56171 → 8801 Len=164
13	0.072464	134.224.131.50	192.168.1.2	UDP	69	8801 → 56172 Len=27
14	0.082008	192.168.1.2	134.224.131.50	UDP	186	56171 → 8801 Len=144
15	0.103554	192.168.1.2	134.224.131.50	UDP	392	56172 → 8801 Len=350
16	0.103839	192.168.1.2	134.224.131.50	UDP	198	56171 → 8801 Len=156
17	0.118939	134.224.131.50	192.168.1.2	UDP	69	8801 → 56172 Len=27

3. In year 2022, the "Digital Marketing Ltd" company's SDN based network suffered from a DOS attack making staff and clients unable to connect to the company network. Staff was not able to access their emails and other company resources, while clients were not able to access the e-market online buying platform. The "Digital Marketing Ltd" company lost over USD$15 million of sales during the attack.

   3.1. Assuming that the above server was utilized TCP as the transport layer protocol, Use a diagram an discuss how TCP is Vulnerable to DoS attacks. You have to clearly show the TCP Queues, and TCP Messages used to give a DoS attack. **(6 marks)**

   3.2. The Below Figure illustrates the Defense in depth. Select the best two layers can be used eliminate the DOS attack. You must compare those with examples. **(6 Marks)**

